



Oxygen Cylinder Safety

Do ✓

Do not ✗

DO LEARN PROPER MEDICAL CYLINDER'S SAFETY HANDLING

- Read and follow the cylinder labeling instructions.



DO NOT ALTER, TRANSPORT OR HANDLE CYLINDERS INCORRECTLY

- Do not change the labeling or repair a cylinder.
- Do not transport gas cylinders in the passenger compartment of a vehicle.
- Do not handle more than one cylinder at a time, or roll cylinders along the ground, except on carts designed for handling gas cylinder.



DO TRANSPORT CYLINDERS CORRECTLY

- Use personal protective equipment and mechanical assistance when handling cylinders (e.g. trolleys).
- Ensure cylinder (regardless of size) is firmly secured by a strong chain or strap, capable of preventing the cylinder from falling or being knocked over.
- Ensure valve guards or caps are fitted when cylinders are not in use or when being transported for delivery.



DO NOT USE UN-CERTIFIED MEDICAL OXYGEN CYLINDERS

- Do not refill cylinders that are not meant for medical oxygen (e.g. cylinders used for other industrial gases) and that have not passed a quality test by a specialist.



DO SET UP CYLINDERS FOR CLINICAL USE AT A SAFE DISTANCE FROM THE PATIENT

- Ensure that the gas is only turned on when is required. Adequate valves, pressure regulators and flowmeters should be placed to control the desired rates.
- Oxygen cylinder valves should be opened smoothly to avoid (adiabatic) compression and heat generation and associated fire risks.
- Ensure adequate ventilation on the wards to reduce the risk of fire.



DO NOT USE OIL, LUBRICANTS OR ALCOHOL-BASED HAND SANITIZER ON CYLINDER'S FITTINGS



DO STORE CYLINDERS CORRECTLY

- Always physically separate full and empty medical cylinders.
- Store all oxygen cylinders in upright position and nesting, shall have three points of contact.
- Ensure that the storage room is well ventilated, clean and not exposed to extremes of temperature and humidity.
- Keep oxygen sources several metres from ignition sources (for example, acetylene used in maintenance).
- Ensure appropriate fire extinguishers are kept nearby and are regularly inspected.



DO NOT ATTEMPT TO REPAIR A CYLINDER OR A VALVE IF LEAKAGE IS DETECTED

- Replacement of damaged components is suggested.



Sources:
WHO/UNICEF Technical Specifications and Guidance for oxygen therapy devices. <https://www.who.int/publications/m/item/oxygen-therapy-devices-2019>
GOV.UK Department of Health, Medical gases - Health Technical Memorandum 52-01: Medical gas pipeline systems. Part 6: Operational Management. Department of Health. London, House of Commons Library, 2006. https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/240006/h52-01-part-6.pdf

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Even hospital personnel are not always aware of how **dangerous** compressed gasses can become.

Remember the following points:

- Oxygen in a cylinder is under high pressure. In addition, oxygen supports combustion, which means it will cause an explosion if mishandled.
- Never let tanks stand upright without a holder because they might fall over, causing the valve to break off. It becomes a 100 lb. "balloon" missile flying around with the potential to inflict serious harm.
- **Always put them in a cage, cart, or holder.** All carts must have cylinders secured with thumb screw or other tether device.
- Only 12 tanks per smoke compartment (excluding tanks in use or if indicated).



FULL:

Tank has the black cap on and gauge indicator is in the green



PARTIAL

No black cap and gauge indicator **not** in the red - more than 1/4 full)



EMPTY:

Gauge indicator in the red - less than 1/4 full)

